

POSTS AND TELECOMMUNICATIONS INSTITUTE OF TECHNOLOGY
FACULTY OF INFORMATION TECHNOLOGY 1



Topic: Garage management

Class:

E22HTTT

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Module 1: Customer Book An Appointment Online
Module 2: Management Staff Views Statistics Of Supplier

Subject No. 31

Duration: 90 minutes

A garage management system enables management staff, sales staff, warehouse staff, and customers to use:

- Management staff: view the statistics categories: services, parts, customers and suppliers. Service information management, spare parts.
- Warehouse staff: import spare parts from suppliers, manage supplier information
- Sales staff: receive customers, receive service and spare parts requests from customers, assign technical staff according to customer service requirements, receive payment from customers.
- Customer: Search service information, book an appointment online

Considering two modules and answering questions

- ***Customer books an appointment online***: selects the appointment booking menu → chooses a date and time → enters the customer's personal information and confirms → the system saves to the database.
- ***Management staff views statistics of supplier***: selects the menu to view the statistical reports → selects the supplier statistics by imported quantity → selects the start and end date of the statistics → views the supplier statistics → clicks on a supplier → view the details of the supplier's import invoices → clicks on an invoice → views the selected imported invoice.

I. CONCEPT EXPLORATION

GARAGE MANAGEMENT SYSTEM GLOSSARY LIST

NO.	English Concept	Meaning
Concepts related to Humans (Actors)		
1	Management Staff	An authorized user responsible for overseeing garage operations, viewing analytics, and managing master data such as services and spare parts.
2	Warehouse Staff	A user role tasked with managing inventory, including the procurement of spare parts from suppliers and maintaining supplier records.
3	Sales Staff	The primary point of contact for customers. This role handles customer reception, documents service requests, assigns technicians, and processes payments.
4	Technical staff	A skilled staff member who performs the maintenance and repair services on customer vehicles.
5	Customer	An individual who utilizes the garage's services. Customers can search for service information and schedule appointments via the system.
6	Supplier	An external entity or company that provides spare parts and other materials to the garage.
Concepts related to Objects (Entities)		
7	Service	A defined maintenance or repair task offered by the garage (e.g., oil change, brake inspection, engine diagnostics).
8	Spare Part	An individual inventory item or component used for vehicle repairs and services.
9	Appointment	A scheduled booking made by a customer for a specific service at a designated date and time.

10	Import Invoice	A financial document that records the transaction details of spare parts procured from a supplier.
11	Statistics Report	A system-generated summary of operational data, providing insights into business performance across various metrics.
Concepts related to Activities (Processes)		
12	Schedule Appointment	The process by which a customer books a future service slot through the online portal.
13	Search Service Information	The functionality allows customers to look up details, descriptions, and pricing of available services.
14	Procure Spare Parts	The workflow for warehouse staff to register the receipt of new inventory from a supplier.
15	View Statistics	The function enables management staff to access and analyze performance reports and data dashboards.
16	Assign Technician	The task performed by a Sales Staff to allocate a specific technician to a customer's service request.
17	Process Payment	The final step in the service workflow where the Service Advisor finalizes the bill and receives payment from the customer.

II. BUSINESS MODEL

A. Requirements:

1. By Natural Language:

1.1. Objective and Scope

- **Objective:** The primary objective of this system is to deliver a robust, centralized software solution to streamline and manage the core operations of an automotive garage. This includes service management, inventory control, supplier relations, customer interactions, and performance analytics.

- **Scope:**
 - **Application Type:** Web-based platform accessible to both internal staff and external customers.
 - **In-Scope Actors:**
 - Customer: The person who use the services, can search for information on services and spare parts, and schedule appointments online.
 - Management Staff: The person responsible for overall management, monitoring, and evaluating activities related to services, spare parts, customers, and suppliers.
 - Warehouse staff: The person responsible for inventory management, receiving and dispatching spare parts from suppliers.
 - Sales staff: The person who receives customers, records service and spare part requests, assigns tasks to technical staff, and handles payments.
 - **In-Scope Functionalities:**
 - The customer schedules an appointment online.
 - Management Staff views supplier statistics.
 - The customer searches for service and parts information.
 - Warehouse Staff imports spare parts.
 - Sales Staff creates a new repair order.
 - Sales Staff processes payment for repair services.
 - Management Staff views service and parts revenue statistics.
 - Management Staff views customer revenue statistics.

1.2. Detailed Business Process of Functions (How do modules work)

- **Module 1: Customer Schedules an Appointment Online**
 1. The customer navigates to the "Schedule Appointment" section from the main menu.
 2. The system presents an interface for the customer to select a desired date and time slot.
 3. The customer populates a form with their personal information (name, contact number) and vehicle details, then submits the request.

4. The system performs validation on the submitted data. If any information is invalid (e.g., missing fields, a past date is selected), the system prompts the user for correction.
5. Upon successful validation, the system saves the appointment record to the database and displays a confirmation message to the customer.

- **Module 2: Management Staff Views Supplier Statistics**

1. The manager logs into the system with their credentials.
2. The manager navigates to Menu → Statistical Reports → Supplier Statistics by Import Quantity.
3. The system displays a form requiring the manager to specify a start date and an end date for the reporting period.
4. The system validates the date range. If the range is invalid (for example: start date is after end date, a date is missing), the system returns an error and requests new input.
5. For a valid date range, the system queries the database and renders a report listing all suppliers, sorted by the total quantity of parts supplied within that period.
6. The manager clicks on a specific supplier from the list. The system then displays all import invoices associated with that supplier for the selected period.
7. The manager selects a specific invoice to view its detailed information, including the invoice ID, date, a list of included spare parts, quantities, and total cost.

- **Module 3: Customer Searches for Service Information**

1. The customer navigates to the "Service & Parts Information" section from the main menu.
2. The customer enters a keyword (service or part name) into the search bar and initiates the search.
3. The system queries the database and returns a list of services and/or parts that match the keyword.
4. If no results are found, the system displays a "No results found" notification and prompts the user to try a new search.
5. If results are available, the customer selects a specific item from the results list.

6. The system renders a detailed information page for the selected item, displaying its description, specifications, pricing, and other relevant data.

- **Module 4: Warehouse Staff Imports Parts**

1. The Warehouse Staff logs into the system and navigates to the "Procure Parts" or "New Inventory" module.
2. The staff searches for a supplier by name or ID.
3. If the supplier record does not exist, the system provides an option to "Add New Supplier," where the staff populates the required fields and saves the new record.
4. The staff selects the appropriate supplier to create a new import invoice.
5. The staff repeats the following sub-process for each part to be procured: a. Search for the spare part by name or part number. b. If the part is new to the inventory, an option is provided to "Add New Part" to the master list. c. Select the part and enter the quantity and unit price. d. Add the item to the current invoice line items.
6. After adding all items, the staff reviews and confirms the final invoice.
7. The system persists the invoice record, automatically updates the inventory stock levels for the procured parts, and generates a unique invoice ID.
8. The system provides an option to print the import invoice for record-keeping and payment processing.

- **Module 5: Sales Staff Receives Car for Repair**

1. The Sales Staff logs into the system and initiates the "New Repair Order" process.
2. The Sales Staff searches for the customer by name, phone number, or license plate.
3. If the customer is new, the system prompts the Sales Staff to create a new customer profile, capturing their personal and vehicle information.
4. Upon selecting the customer, the system generates a new, blank Repair Order.
5. The Sales Staff adds all required services and parts to the Repair Order by searching for and selecting each item.

6. After itemizing all services and parts, the Sales Staff assigns an available technician to the job from a system-provided list.
7. The Sales Staff confirms the final details of the Repair Order with the customer and finalizes it.
8. The system saves the Repair Order, assigns it a unique ID, and updates the vehicle's status to "In Service."

- **Module 6: Sales Staff Receives Payment**

1. The Sales Staff navigates to the "Process Payment" module.
2. The Sales Staff retrieves the customer's account by searching their name or vehicle information.
3. The system displays a list of completed and outstanding (unpaid) invoices for that customer.
4. The Sales Staff selects the specific invoice to be paid.
5. The system renders the detailed invoice, itemizing all services, parts, and the final amount due.
6. If authorized, the Sales Staff can make last-minute adjustments or add items to the invoice before finalizing.
7. The Sales Staff confirms the final amount with the customer and proceeds to process the payment.
8. The Sales Staff records the payment method (cash, card, etc.) and amount in the system.
9. The system updates the invoice status to "Paid" and closes the Repair Order.
10. The system generates a final receipt and provides an option to print it for the customer.

- **Module 7: Management Staff Views Service/Parts Statistics**

1. The Management Staff logs into the system.
2. They navigate to Menu → Statistical Reports → Service/Parts Statistics by Revenue.
3. The manager specifies a start date and an end date for the reporting period.
4. If the date range is invalid (e.g., start date is after end date), the system will return an error and prompt for correction.
5. The system queries the data and renders a report of services and parts, sorted by total revenue within the specified period.

6. The manager can drill down by clicking on a specific service or part, which displays a list of all associated invoices within the period.
 7. Selecting an invoice from this list displays its full details.
- **Module 8: Management Staff Views Customer Statistics**
 1. The Management Staff logs into the system.
 2. They navigate to Menu → Statistical Reports → Customer Statistics by Revenue.
 3. The manager specifies a start date and an end date for the reporting period.
 4. If the date range is invalid, the system will return an error and prompt for correction.
 5. The system queries the data and renders a report of customers, sorted by their total revenue generated within the period.
 6. The manager can drill down by selecting a specific customer, which displays a list of that customer's invoices from the period.
 7. Further selecting an invoice from the list reveals its complete details.

1.3. Information about Related Objects (Data Entities)

- **Staff:** staffID, name, contactInfo, username, hashedPassword, role ('Manager', 'Sales', 'Warehouse', 'Technical')
- **Vehicle:** licensePlate, make, model, year, ownerCustomerID (foreign key).
- **Appointment:** appointmentID, customerDetails, vehicleInfo, serviceType, appointmentDate, appointmentTime, status.
- **Supplier:** supplierID, supplierName, contactInformation, address.
- **ImportInvoice:** invoiceID, supplierID, issueDate, lineItems (list of parts), totalQuantity, totalValue.
- **Customer:** customerID, name, contactInfo, vehicleHistory.
- **SupplierStatisticsReport:** supplierID, supplierName, totalImportQuantity.
- **Service:** serviceID, serviceName, description, price.
- **SparePart:** partID, partName, description, price, quantityInStock.

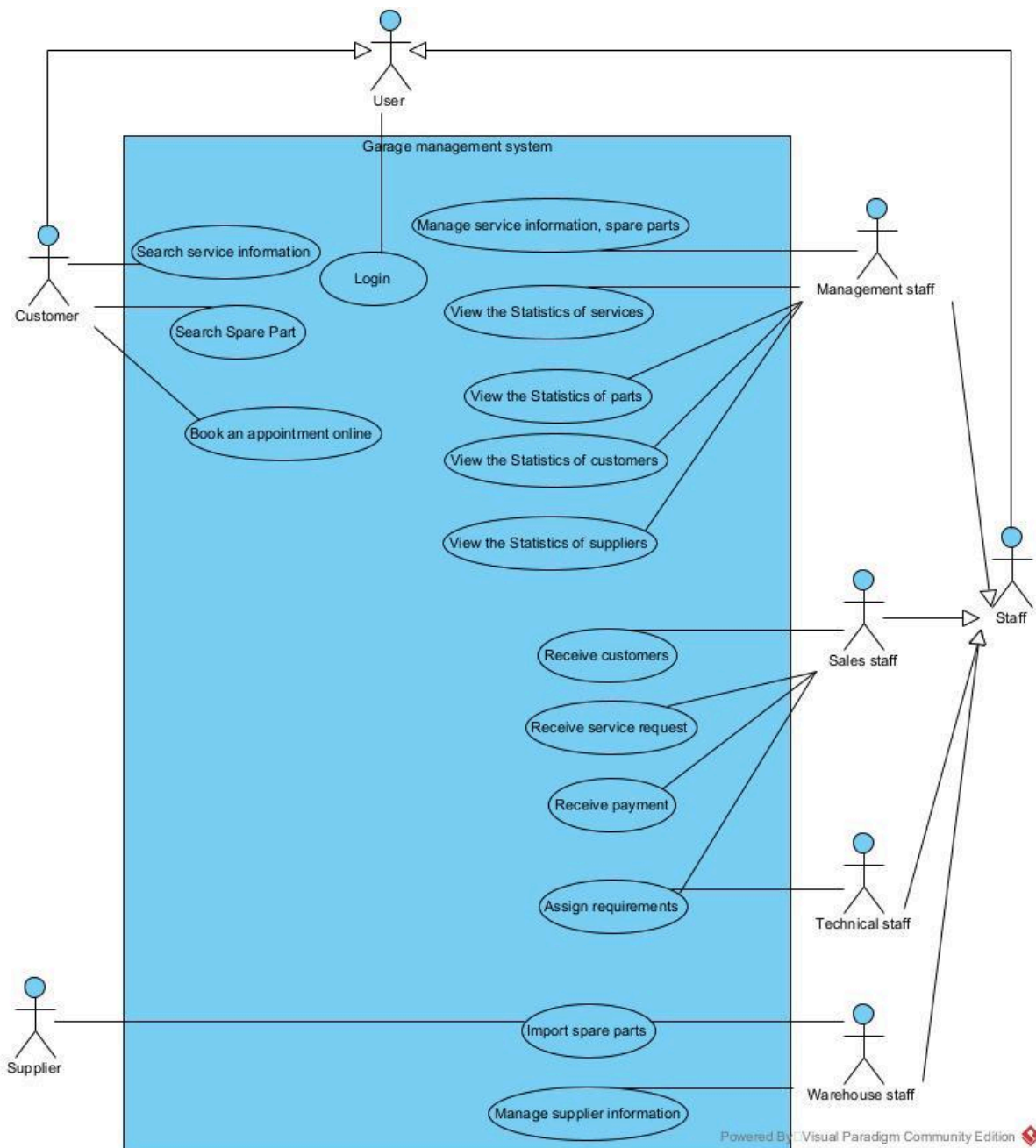
- **RepairOrder:** orderID, customerID, vehicleLicensePlate, creationDate, serviceAdvisorStaffID, assignedTechnicianStaffID, lineItems (containing serviceID or partID), totalAmount, status.

1.4. Relationships among Objects

- A **Customer** has a one-to-many relationship with **Vehicle** (one customer can own multiple vehicles).
- A **Customer** has a one-to-many relationship with **RepairOrder** (one customer can have multiple repair orders over time).
- A **Customer** has a one-to-many relationship with **Appointment** (one customer can schedule multiple appointments).
- A **RepairOrder** has a many-to-one relationship with **Staff** (each repair order is created by one *Sales Staff* and assigned to one *Technical Staff*).
- A **RepairOrder** has a many-to-many relationship with **Service** and **SparePart** (a repair order can contain multiple services and parts, and a specific service or part can appear on many different repair orders).
- A **Supplier** has a one-to-many relationship with **ImportInvoice** (one supplier can issue multiple import invoices).
- An **ImportInvoice** has a many-to-many relationship with **Spare Part** (an invoice can list multiple parts, and a specific part can be included in multiple invoices).

2. By UML:

2.1. General Use Case



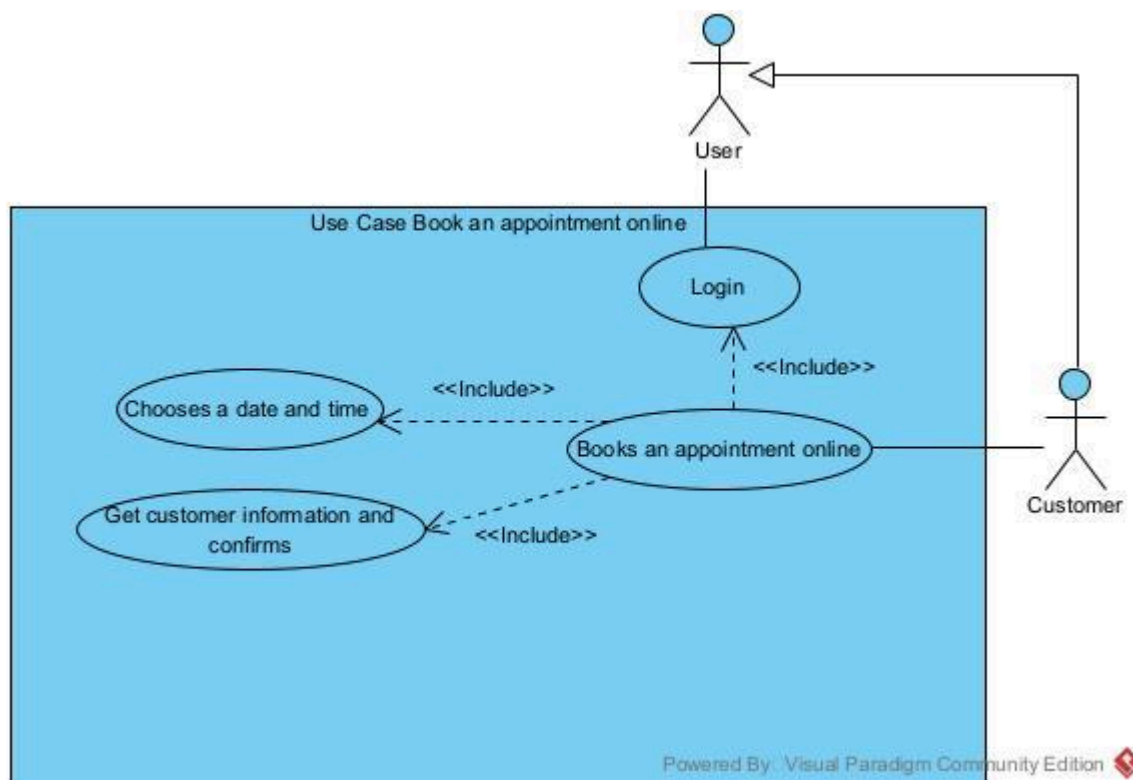
Description for General Use Case:

- Search service information: This use case enables customers to search for services and spare parts using keywords or IDs.
- Book an appointment online: This use case allows customers to book appointments online at a specific time, provided the slot is valid.
- Login: This use case enables user authentication.
- Manage service information, spare parts: This use case allows management staff to manage and update information about services and spare parts.

- Receive customers: This use case allows sales staff to receive incoming customers and verify or create their profiles in the system.
- Receive service request: This use case enables sales staff to collect and record customer service requests, including selected services and spare parts.
- Receive payment: This use case enables sales staff to process payments for completed services and generate receipts.
- Assign requirements: This use case allows sales staff to assign tasks and requirements to available technicians based on the repair order.
- Import spare parts: This use case allows warehouse staff to import spare parts into inventory, create import invoices.
- Manage supplier information: This use case enables warehouse staff to manage supplier records, including creating, updating, and maintaining supplier data.

2.2. Module 1: Customer Schedules an Appointment Online

Use Case books an appointment online:

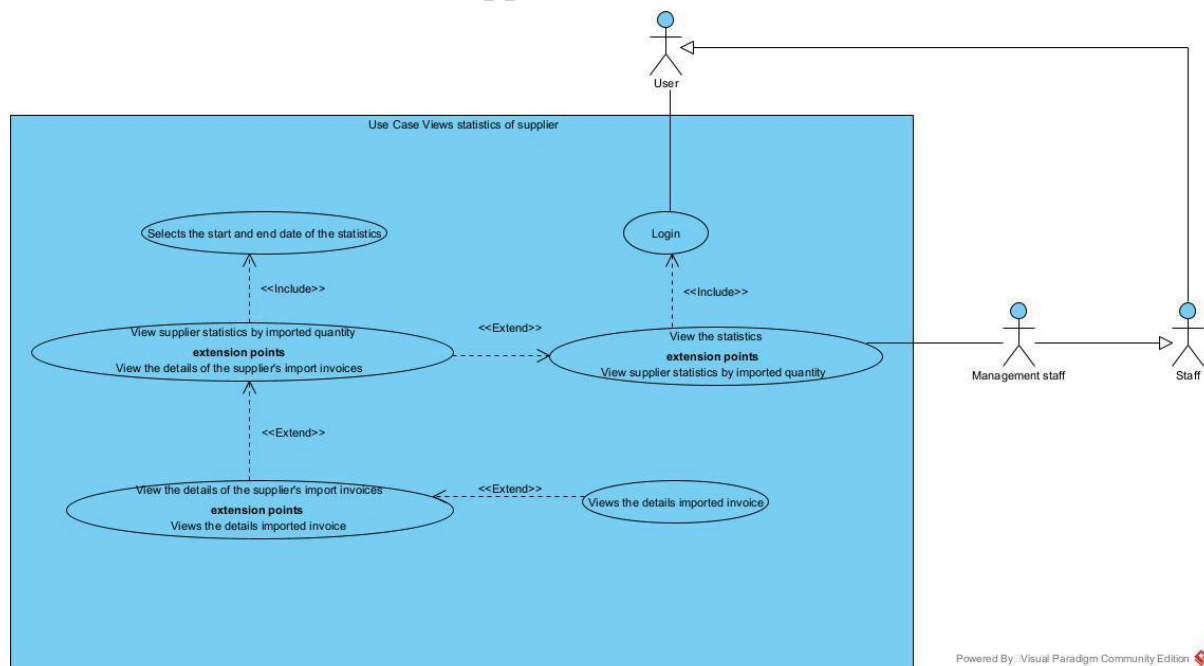


Description for Use Case books an appointment online:

- Book an appointment online: This use case enables the customer to schedule a service appointment for their vehicle at a specific date and time through the online system.
- Chooses a date and time: This use case enables the customer to choose a valid date and time for setting the appointment
- Get customer information and confirm: This use case enables the customer to inform personal data.

2.3. Module 2: Management Staff Views Supplier Statistics

Use Case Views statistics of supplier



Description for Use Case Views statistics of supplier:

- View the statistics: This use case enables the Management Staff to generate and view statistical reports.
- View supplier statistics by imported quantity: This use case enables the Management Staff to view statistics about the supplier by imported spare parts quantity.
- Selects the start and end date of the statistics: This use case enables the Management Staff to choose a start date and end date to fill out the time needed.
- View the details of supplier's import invoices: This use case enables the Management Staff to view the list of import invoices.
- Views the details of imported invoices: This use case enables the Management Staff to view the details of imported invoices.

B. Analysis:

1. Scenario:

Module 1: Customer Schedules an Appointment Online

Use case	Customer Schedules an Appointment Online
Actor(s)	Customer
Preconditions	The customer has an existing account and is accessing the garage's website or application. The system has available time slots in its schedule.
Postconditions	Success: A new appointment record is created and saved in the system. The chosen time slot is reserved. The customer receives a confirmation. Failure: No new appointment is created. The system state remains unchanged.
Main events	1. From the homepage (after being logged in), Customer A selects the "Book an Appointment" function. 2. The system navigates to the scheduling interface, displaying a table with available dates. 3. Customer A selects a desired date, for example: September 25, 2025. 4. The system displays a list of available time slots for the selected date, for example: 9:00 AM, 10:00 AM, 2:00 PM. 5. Customer A selects the "9:00 AM" time slot. 6. The system displays a form, automatically filling it with the customer's saved information (Name, Phone Number, Vehicle) since the customer is already logged in. 7. Customer A reviews the information, fills in the text box to describe the vehicle's problems, and clicks the "Confirm" button. 8. The system validates the data, saves the appointment, displays a "Successfully Scheduled!" message, and returns to the homepage.
Exceptional scenarios	3. Login Failure: At step 3, if the customer enters an incorrect username or password, the system displays an

	error message. 3.1. The customer clicks the OK button of the error message. 3.2. The system returns to the login UI. 7. No Available Slots/Slot Taken: At step 7 or 10, the time slot the customer selected is booked by another user just before the customer confirms. The system will display an error message and ask the customer to select a different time slot. 10. Invalid Form Submission: If the form data is incomplete or invalid (e.g., missing required service), the system will display an error highlighting the fields that need correction. 10.1. The customer fill in the highlight fields and clicks the "Confirm Appointment" button again. 10.2. The system validates the data, saves the appointment, and displays a "Successfully Scheduled!" message with the appointment details.
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Module 2: Management Staff Views Supplier Statistics

Use case	Management Staff Views Supplier Statistics
Actor(s)	Management Staff
Preconditions	The management staff has an existing account and is accessing the garage's website or application. The system contains historical data of import invoices from suppliers.
Postconditions	Success: The system successfully displays the statistical report to the manager. The underlying system data is not altered. Failure: The report is not displayed. The system state remains unchanged.
Main events	1. From the homepage, management staff selects the "Reports" function. 2. The system prompts the management staff to log in to proceed. 3. Management staff enters their username and password,

	then clicks the "Log In" button. 4. The system successfully authenticates the user and navigates to the Reports interface, displaying several options which containing "Supplier Statistics" option. 5. Management staff click on the "Supplier Statistics" option. 6. The system immediately displays a report interface, which includes: • Date range selectors at the top, defaulted to the last 30 days (e.g., from August 15, 2025, to September 14, 2025). • Below the selectors, a table listing suppliers, ranked by the total quantity of parts imported during that default period. 7. The manager wants to view data for a different period, so they change the start date to July 1, 2025, and the end date to July 31, 2025. After that, the manager clicks on the "Apply" button. 8. The system displays the supplier list table below immediately refreshes to display the statistical data for July. 9. The manager wants to see more details for the top supplier, "XYZ Parts Corp," and clicks on the row of the company's name. 10. The system displays a list invoice UI, and a list of all import invoices from "XYZ Parts Corp" within July appeared. 11. The manager clicks on a specific row of invoice in order to view the details of the invoice, for example, "INV-0001," to view its details. 12. The system displays the full details of invoice "INV-0001." 13. After viewing the details invoice the manager clicks on the "Return" button. 14. The system displays a list invoice UI. 15. The manager is also done with the list of invoices so continue to click on the "Return" button. 16. The system displays the report interface. 17. The manager clicks on the "Return" button one more time in order to come back to the homepage. 18. The system displays the homepage.
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Exceptional scenarios	3. Login Failure: At step 3, if the customer enters an incorrect username or password, the system displays an error message. 3.1. The customer clicks the OK button of the error message. 3.2. The system returns to the login UI. 6. Invalid Date Range: At step 6, if the manager enters a start date that is after the end date, the system will display an error message: "Start date cannot be after the end date." 6.1. The customer clicks the OK button of the error message and fixes the start date and end date. 8. No Data Available: At step 8, if there are no import invoices within the selected period, the system will display a message: "No data is available for this period." 8.1. The customer clicks the OK button of the error message 8.2. The system returns to the report interface.
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2. Entity classes extraction:

Step 1: Describe the whole system in a complete paragraph

The garage management system is a comprehensive software solution designed to streamline the operations of a vehicle repair shop. It serves multiple user roles, including customers, who can search for information on services and spare parts and book appointments online. Internally, the system is used by sales staff to manage the entire customer workflow, from receiving customers and their vehicles for repair, creating service and parts requests, and assigning technicians, to finally processing payments and generating invoices. Warehouse staff utilize the system to manage inventory by importing spare parts from various suppliers and maintaining supplier information. Finally, the management staff has oversight capabilities, allowing them to view detailed statistical reports on services, parts, customers, and suppliers, as well as manage the master data for service and spare part information to ensure operational efficiency.

Step 2: Extract all nouns that appeared in the paragraph

- garage management system
- software solution

- operations
- vehicle repair shop
- user roles
- customers
- information
- services
- spare parts
- appointments
- system
- sales staff
- customer workflow
- vehicles
- repair
- service requests
- parts requests
- technicians
- payments
- invoices
- warehouse staff
- inventory
- suppliers
- supplier information
- management staff
- oversight capabilities
- statistical reports
- master data
- operational efficiency

Step 3: Classify the nouns

a. Nouns that could become an entity:

- **Customer** (from customers)
- **Service** (from services)
- **Spare Part** (from spare parts)
- **Appointment** (from appointments)
- **Staff** (from sales staff, warehouse staff, management staff, technicians)
- **Vehicle** (from vehicles)

- **Repair Order** (from repair, invoices, service requests, parts requests)
- **Supplier** (from suppliers)
- **Import Invoice** (from invoices)

b. Nouns that could become an attribute of an entity class:

- **role** (from user roles) - An attribute of the User entity.
- **payment** (from payments) - Can be represented as a status attribute on the Repair Order entity.
- **supplier information** - Represents the collection of attributes within the Supplier entity (e.g., name, address, contact).

c. Rejected nouns:

- **garage management system, system, software solution:** Too general, refers to the system itself.
- **operations, customer workflow, oversight capabilities, operational efficiency:** Too abstract, describes processes or goals, not concrete objects.
- **vehicle repair shop:** Out of scope, describes the business environment, not a system entity.
- **information, master data:** Too general.
- **inventory:** An abstract concept represented by the Spare Part entity and its quantity attribute.
- **statistical reports:** Derived data, not a primary entity to be stored.

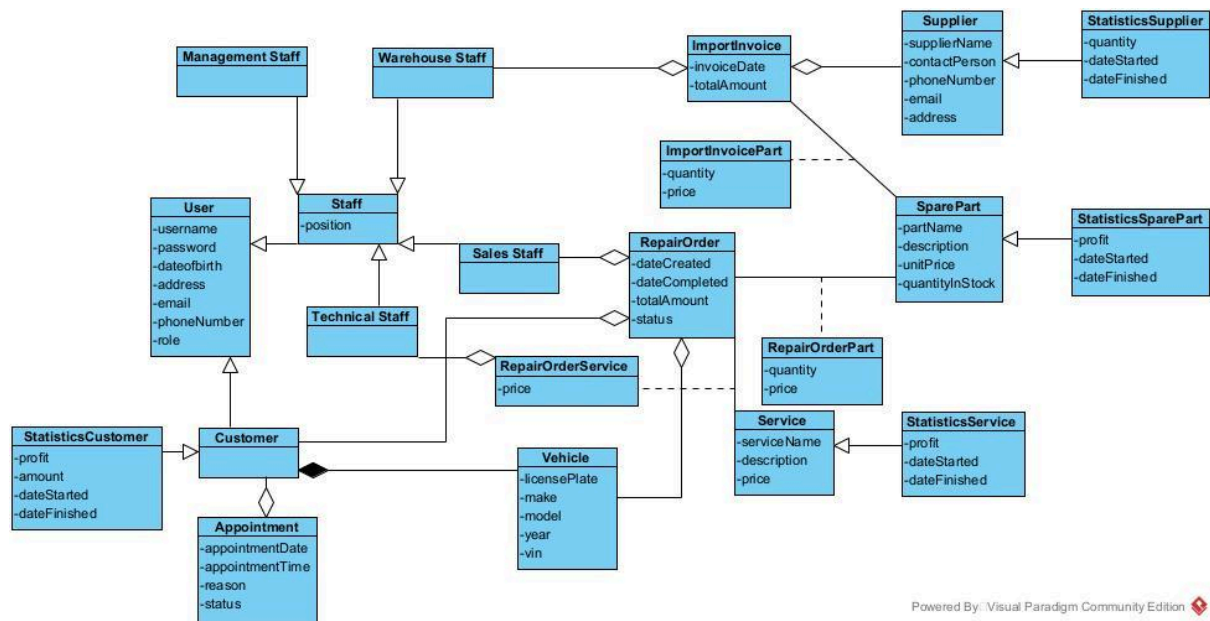
Step 4: Considering the quantitative relationship among entity classes

- **Customer 1 --- N Vehicle:** A customer can own multiple vehicles. **(1-N, Keep)**
- **Customer 1 --- N Repair Order:** A customer can have multiple repair orders. **(1-N, Keep)**
- **Customer 1 --- N Appointment:** A customer can book multiple appointments. **(1-N, Keep)**
- **Staff 1 --- N Repair Order:** One staff member (in a specific role) can be associated with many repair orders. **(1-N, Keep)**
- **Supplier 1 --- N Import Invoice:** A supplier can have many import invoices. **(1-N, Keep)**

- **Repair Order N --- N Service:** A repair order can contain many services and parts, and a service can be on many orders. (N-N) => This is divided into two 1-N relationships by creating a **RepairOrderService** linking entity. A Repair Order has many RepairOrderService, and a Service can be in many **RepairOrderService**.
- **Repair Order N --- N Spare Part:** A repair order can contain many services and parts, and a part can be on many orders. (N-N) => This is divided into two 1-N relationships by creating a **RepairOrderPart** linking entity. A Repair Order has many **RepairOrderPart**, and a Spare Part can be in many **RepairOrderService**.
- **Import Invoice N --- N Spare Part:** An import invoice lists many parts, and a part can be on many import invoices. (N-N) => This is divided into two 1-N relationships by creating an **ImportInvoicePart** linking entity. An Import Invoice has many **ImportInvoicePart**, and a Spare Part can be in many ImportInvoicePart.

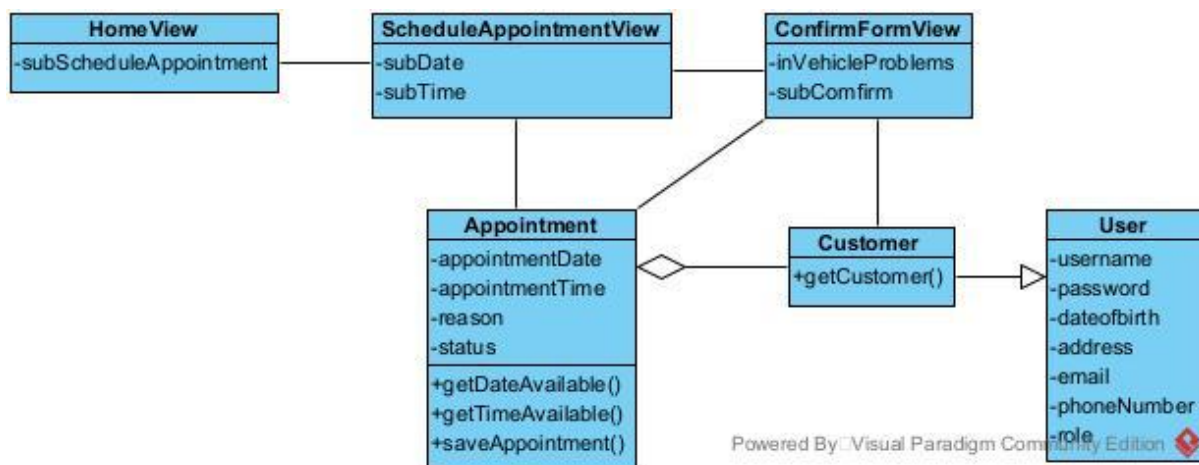
Step 5: Determine the object relationship among entity classes

- **Generalization:** The different staff types (Management Staff, Sales Staff, Technician) can be seen as specializations of a general **Staff** class.
- **Association:** This is the most common relationship, representing a connection between entities.
 - A **Customer** *schedules* an **Appointment**.
 - A **Staff** member *creates* a **Repair Order**.
 - A **Supplier** *issues* an **Import Invoice**.
- **Composition (a strong "whole-part" relationship):**
 - A **Customer** *owns* **Vehicles**. In the context of the system, if a customer record is deleted, their associated vehicle records should likely be deleted as well, indicating a composition relationship.

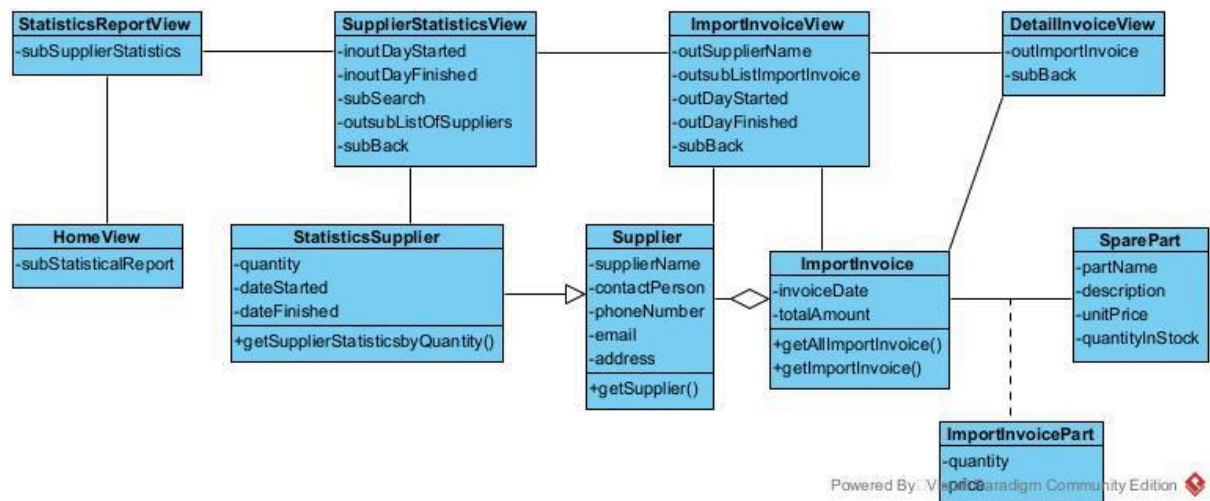


3. Classes diagram and state diagram:

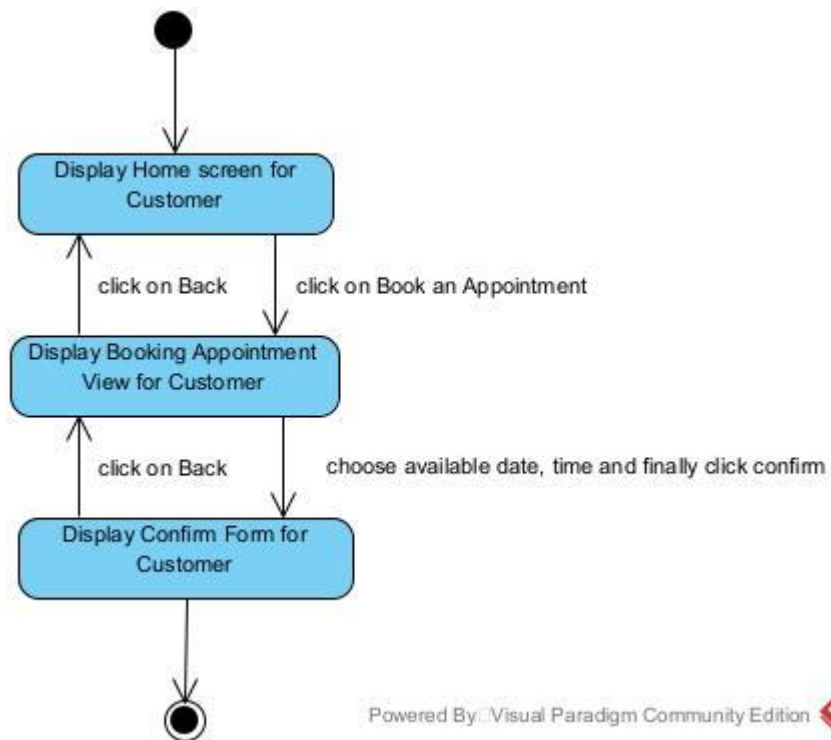
Classes diagram for Booking Appointment Online



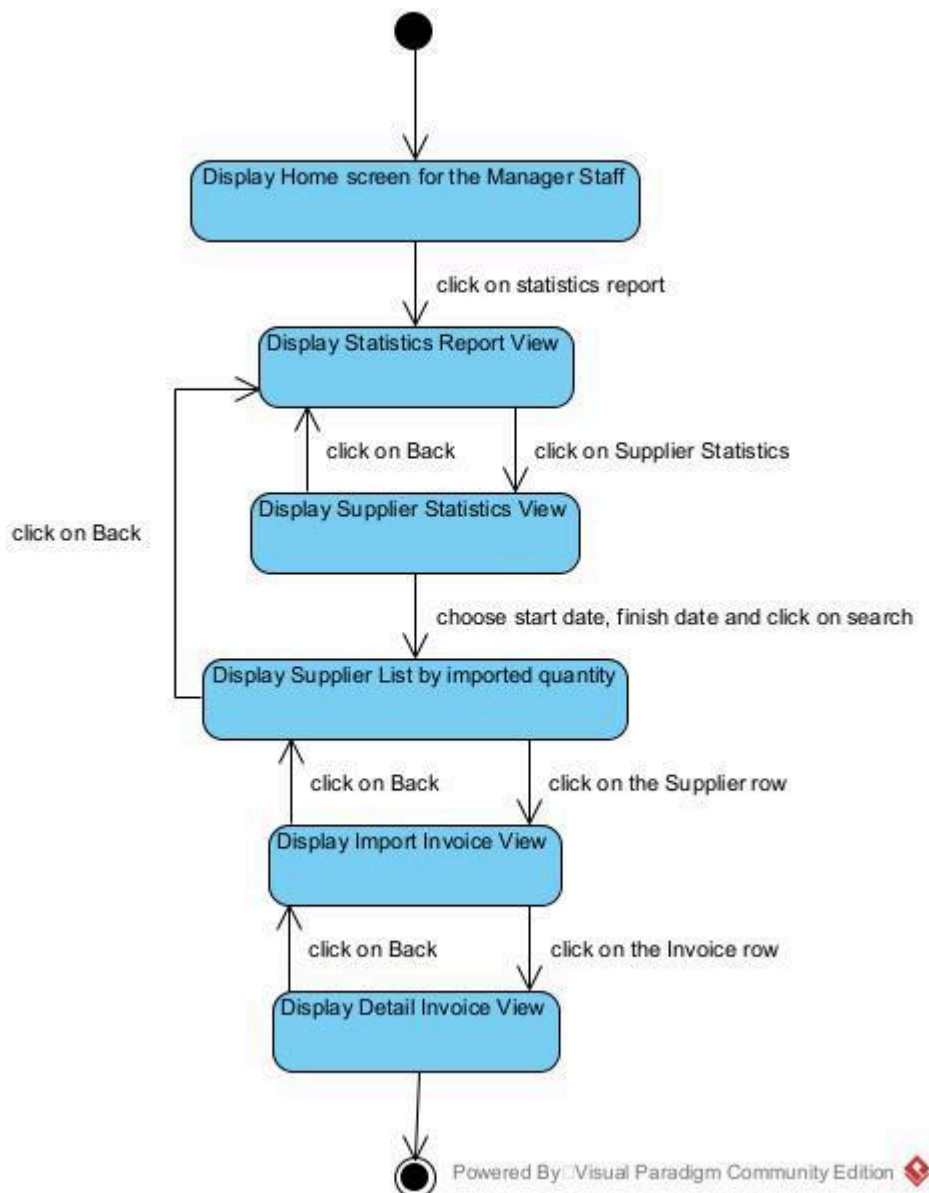
Classes diagram for View Supplier Statistics



State Diagram for Booking Appointment Online



State Diagram for View Supplier Statistics



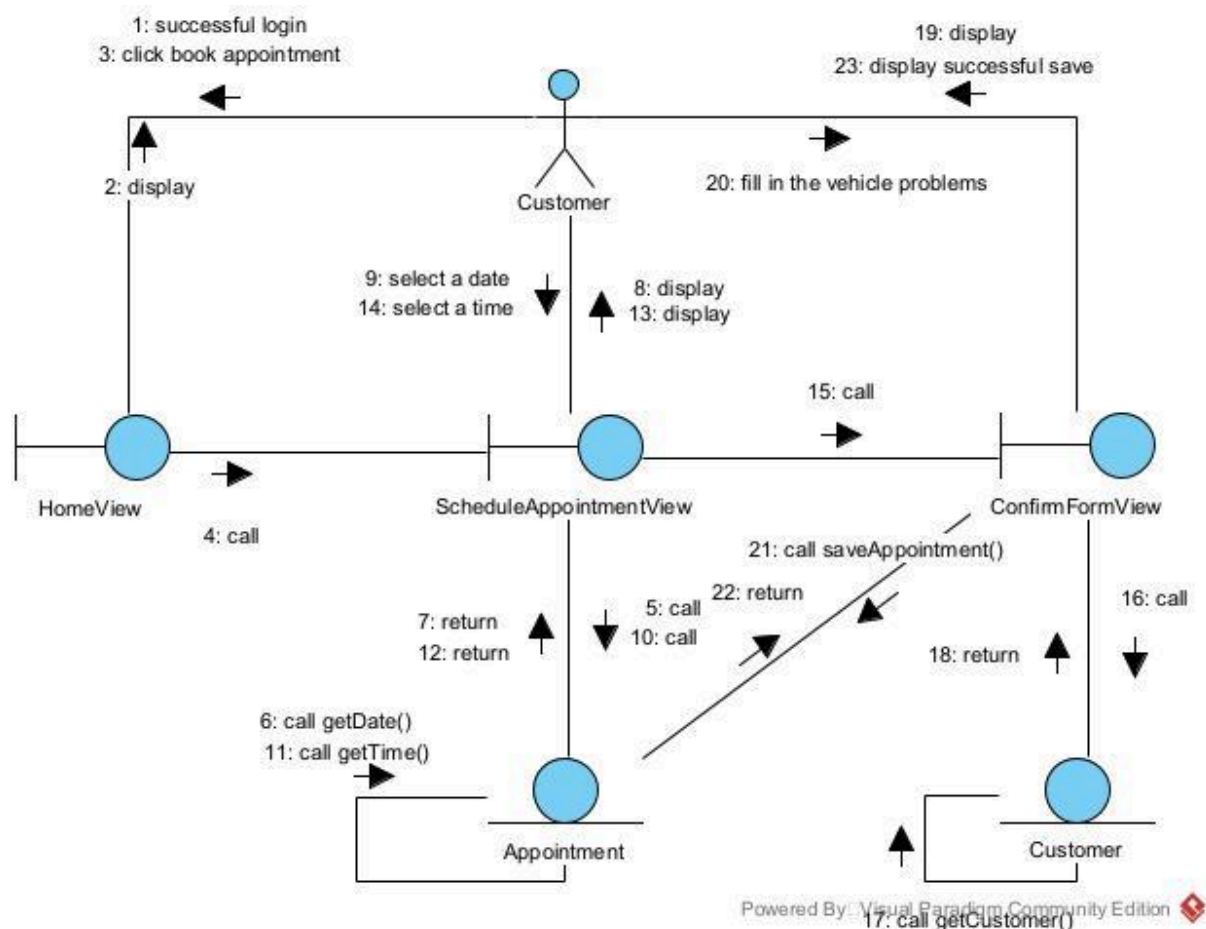
4. Communication diagram and sequence diagram:

Scenario 2 for Booking Appointment online

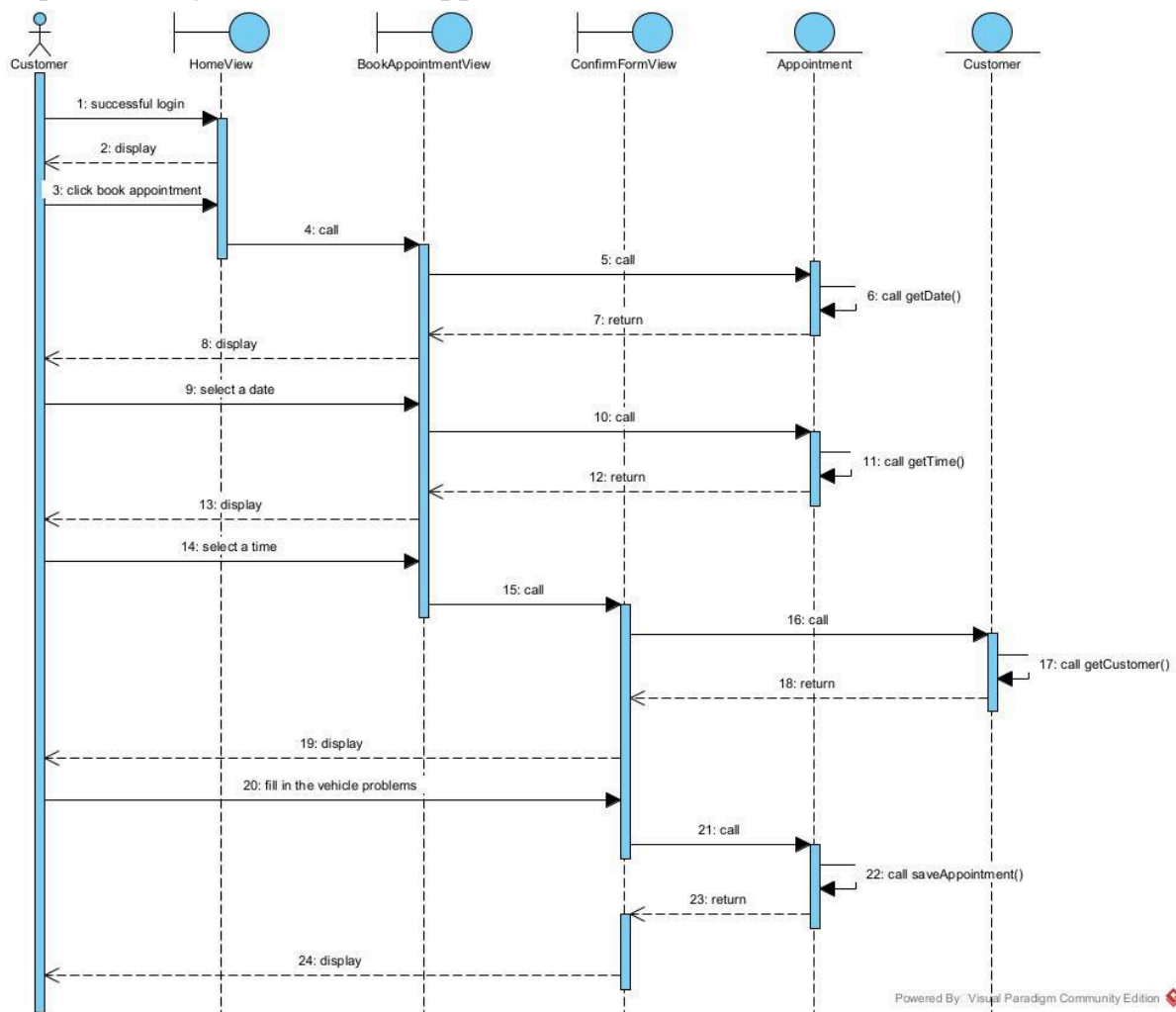
1. The customer successfully logs into the system.
2. HomeView displays the main interface.
3. Customer clicks on Book Appointment.
4. HomeView calls ScheduleAppointmentView to open the appointment form.
5. ScheduleAppointmentView calls Appointment to initialize appointment information.
6. Appointment call getDate() function.
7. Appointment return available date for ScheduleAppointmentView.
8. ScheduleAppointmentView displays the appointment interface to Customer.
9. Customer selects a date.

10. ScheduleAppointmentView calls Appointment to retrieve available times.
11. Appointment call getTime() function.
12. Appointment returns available time to ScheduleAppointmentView.
13. ScheduleAppointmentView displays the available time.
14. The customer selects a time.
15. ScheduleAppointmentView calls ConfirmFormView to prepare the confirmation form.
16. ConfirmFormView calls Customer object to retrieve customer details.
17. Customer call getCustomer() function.
18. Customer return the customer information to ConfirmFormView.
19. ConfirmFormView displays the confirmation form.
20. Customer fills in the vehicle problem information.
21. ConfirmFormView calls Appointment to save Appointment.
22. Appointment call saveAppointment() function
23. Appointment returns a “successful save” message to ConfirmFormView.
24. ConfirmFormView displays the “Save Successful” message.

Communication Diagram for Book Appointment Online



Sequence diagram for Book Appointment Online

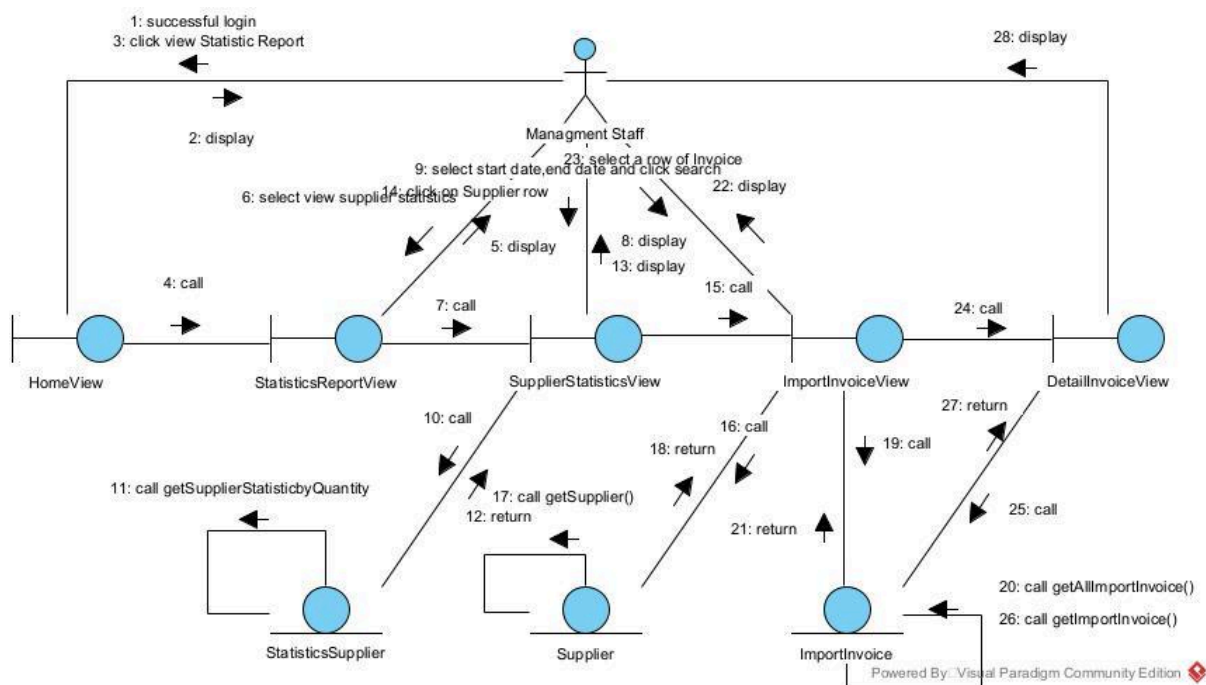


Scenario 2 for Views Supplier Statistics

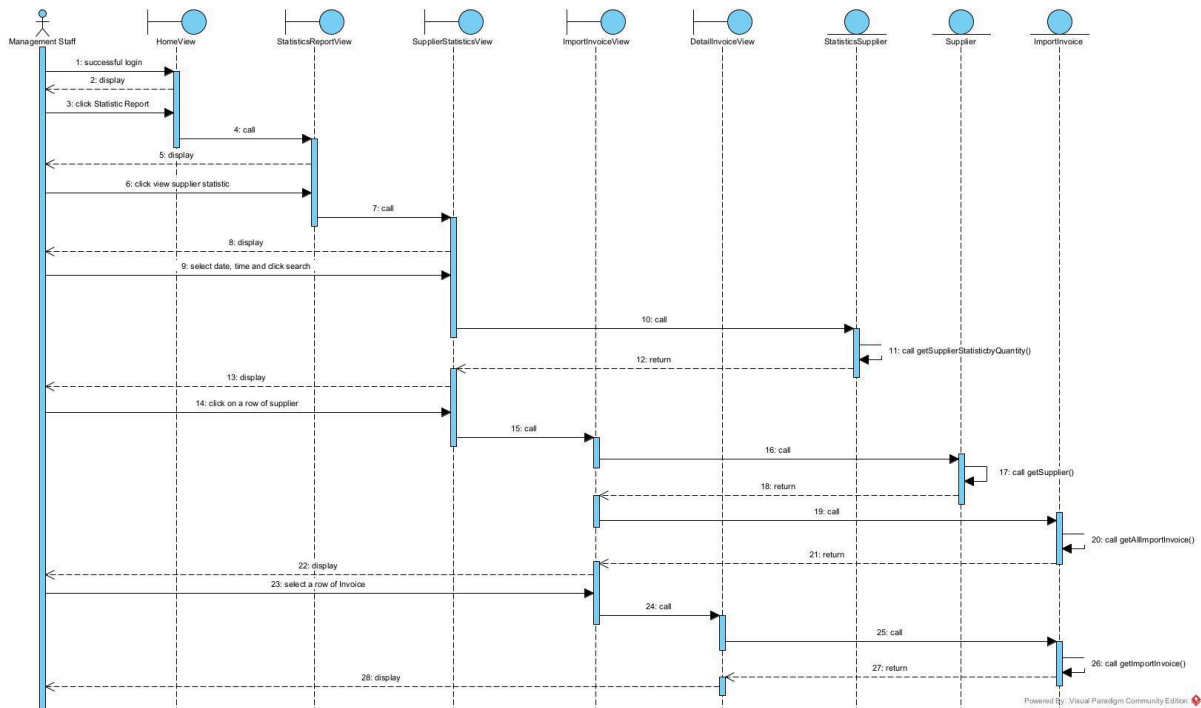
1. Management Staff successfully logs into the system.
2. HomeView displays the main interface.
3. Management Staff clicks on Statistic Report.
4. HomeView calls StatisticsReportView to open the statistics report interface.
5. StatisticsReportView displays the report options.
6. Management Staff clicks View Supplier Statistics.
7. StatisticsReportView calls SupplierStatisticsView.
8. SupplierStatisticsView displays the supplier statistics interface.
9. Management Staff selects date, time, and clicks Search.
10. SupplierStatisticsView calls StatisticSupplier to retrieve supplier statistics.
11. StatisticSupplier calls getSupplierStatisticByQuantity() function.
12. StatisticSupplier returns supplier statistics data to SupplierStatisticsView.

13. SupplierStatisticsView displays the supplier statistics.
14. Management Staff clicks on a row of a supplier.
15. SupplierStatisticsView calls ImportInvoiceView to display related invoices.
16. ImportInvoiceView calls Supplier to get supplier information.
17. Supplier call `getSupplier()` function.
18. Supplier returns data to ImportInvoiceView.
19. ImportInvoiceView calls ImportInvoice to get all import invoices of the supplier.
20. ImportInvoice call `getAllImportInvoice()` function.
21. ImportInvoice returns invoice data to ImportInvoiceView.
22. ImportInvoiceView displays the list of import invoices.
23. Management Staff selects a row of an invoice.
24. ImportInvoiceView calls DetailInvoiceView to display invoice details.
25. DetailInvoiceView calls ImportInvoice to get specific invoice details.
26. ImportInvoice call `getImportInvoice()` function.
27. ImportInvoice returns invoice detail data to DetailInvoiceView.
28. DetailInvoiceView displays the invoice details to Management Staff.

Communication Diagram for Views Supplier Statistics



Sequence diagram for Views Supplier Statistics



C. Design:

1. Entity class diagram:

Step 1:

Classes have been supplemented with the id attribute: except for the statistics classes, the staff classes, and the Customer class

Step 2:

The attributes of the classes are supplemented with attributes according to the Java programming language.

Step 3:

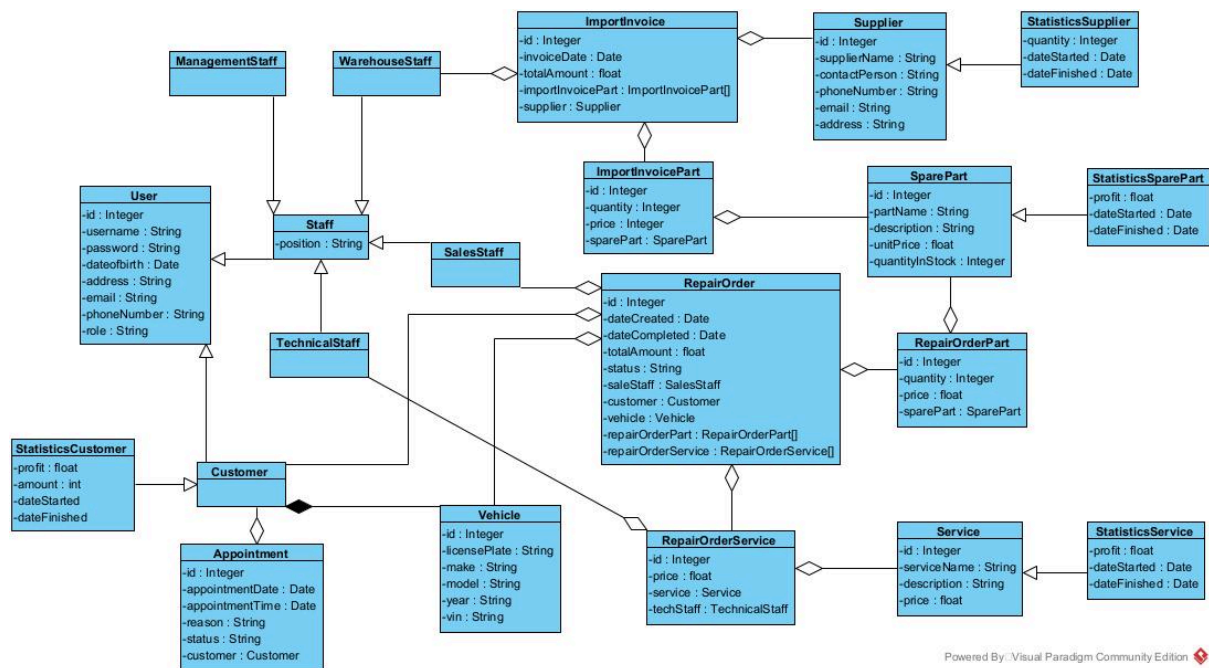
Relation: ImportInvoice - SparePart -> ImportInvoicePart now changed to ImportInvoice contain ImportInvoicePart, ImportInvoicePart contain SparePart.

Relation: RepairOrder - SparePart -> RepairOrderPart now changed to RepairOrder contain RepairOrderPar, RepairOrderPart contain SparePart.

Relation: RepairOrder - Service-> RepairOrderService now changed to RepairOrder contain RepairOrderService, RepairOrderService contain Service.

RepairOrder contains Customer, Vehicle and SalesStaff.

ImportInvoice contains Supplier.



2. Database design:

Step 1: Each entity class suggests a corresponding table

Class User -> tblUser

Class Staff -> tblStaff

Class Vehicle -> tblVehicle

Class Appointment -> tblAppointment

Class RepairOrder -> tblRepairOrder

Class ImportInvoice -> tblImportInvoice

Class Supplier -> tblSupplier

Class ImportInvoicePart -> tblImportInvoicePart

Class SparePart -> tblSparePart

Class RepairOrderPart -> tblRepairOrderPart

Class Service -> tblService

Class RepairOrderService -> tbl RepairOrderService

Step 2: Convert the relation between entity classes into relation between tables:

1 tblUser - n tblAppointment → tblAppointment contains foreign key
tblCustomerid

1 tblUser - n tblVehicle → tblVehicle contains foreign key tblCostumerid

1 tblUser - n tblRepairOrder → tblRepairOrder contains foreign key
tblCustomerid

1 tblVehicle - n tblRepairOrder → tblRepairOrder contains foreign key
tblVehicleid referencing

1 tblRepairOrder - n tblRepairOrderPart → tblRepairOrderPart contains foreign
key tblRepairOrderid

1 tblRepairOrder - n tblRepairOrderService → tblRepairOrderService contains
foreign key tblRepairOrderid

1 tblService - n tblRepairOrderService → tblRepairOrderService contains
foreign key tblServiceid

1 tblSparePart - n tblRepairOrderPart → tblRepairOrderPart contains foreign
key tblSparePartid

1 tblSparePart - n tblImportInvoicePart → tblImportInvoicePart contains
foreign key tblSparePartid

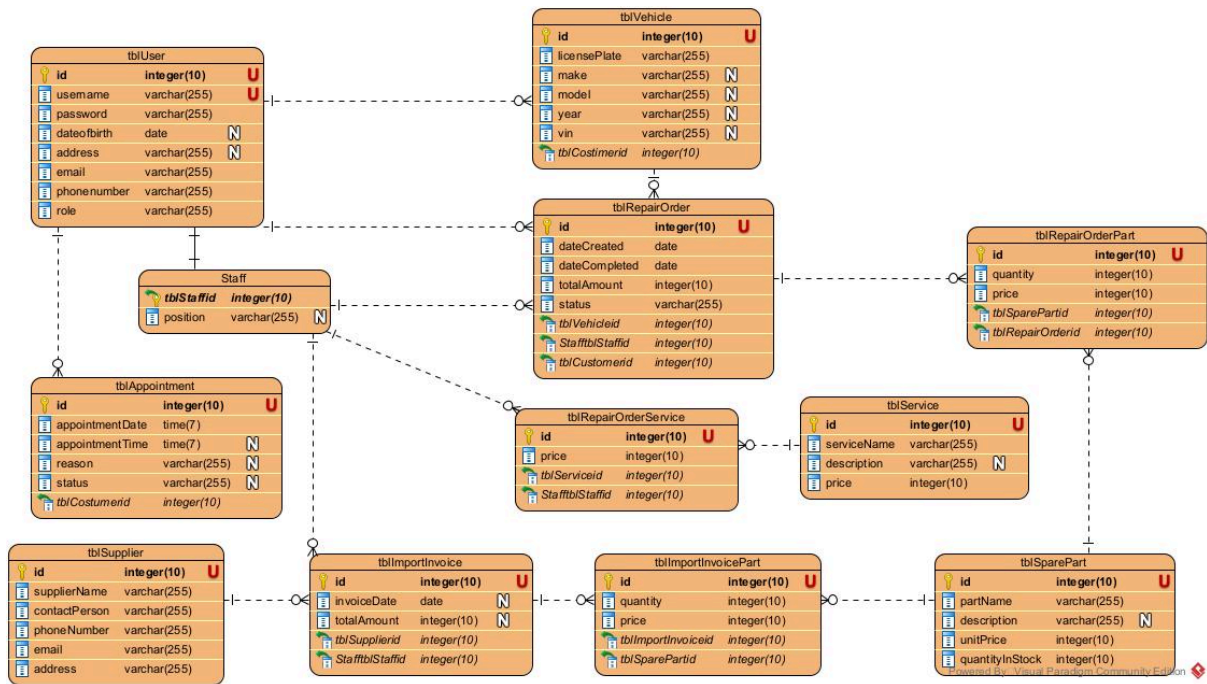
1 tblImportInvoice - n tblImportInvoicePart → tblImportInvoicePart contains
foreign key tblImportInvoiceid

1 tblSupplier - n tblImportInvoice → tblImportInvoice contains foreign key
tblSupplierid

1 Staff - n tblImportInvoice → tblImportInvoice contains foreign key tblStaffid

1 Staff - n tblRepairOrder → tblRepairOrder contains foreign key StafftblStaffid

1 Staff - n tblRepairOrderService → tblRepairOrderService contains foreign
key StafftblStaffid

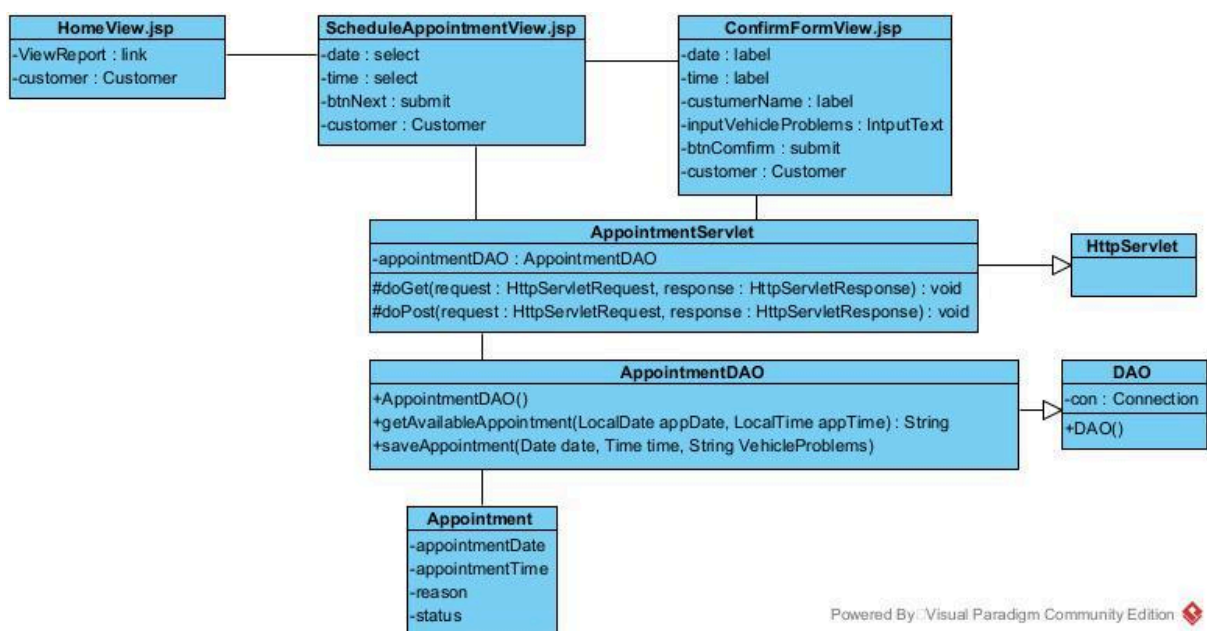


3. Class design diagram:

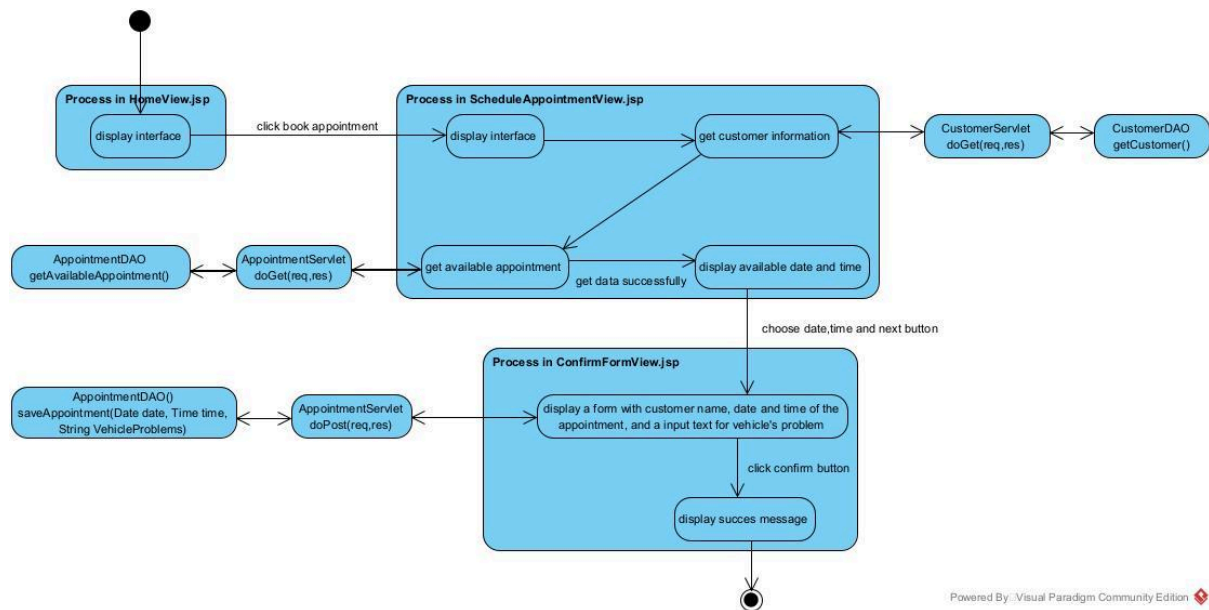
Module 1: Customer books an appointment online

User interface is the jsp page: HomeView.jsp, ScheduleAppointmentView.jsp, ConfirmFormView.jsp.

DAO for manipulate the model: DAO, AppointmentDAO.



Activity Diagram:

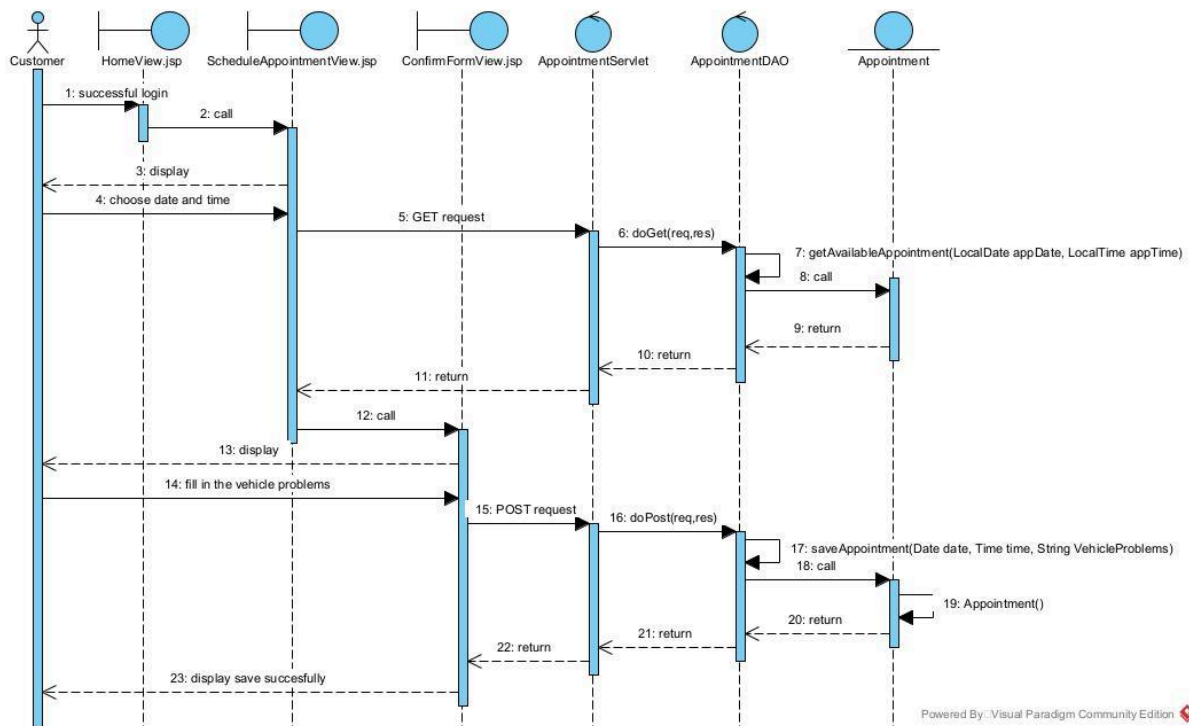


Scenario ver3:

1. On the main interface HomeView.jsp, after successfully logging in, the customer clicks the “Book Appointment” function.
2. The page HomeView.jsp calls the page ScheduleAppointmentView.jsp.
3. The page ScheduleAppointmentView.jsp displays to the customer.
4. The customer chooses the day and time as they want for the appointment and then clicks on the next button.
5. The page ScheduleAppointmentView.jsp sends a GET request to AppointmentServlet to request a list of available appointments.
6. AppointmentServlet receives the request and calls the method doGet(request, response).
7. Inside doGet(), AppointmentServlet calls the class AppointmentDAO to request the list of available appointments.
8. The class AppointmentDAO calls the method getAvailableAppointment().
9. The method getAvailableAppointment() calls the class Appointment.
10. The class Appointment returns the result to the method getAvailableAppointment().
11. The method getAvailableAppointment() returns the result to AppointmentDAO, then to AppointmentServlet.
12. AppointmentServlet calls ConfirmFormView.jsp.
13. The page ConfirmFormView.jsp displays the selected appointment details to the customer.
14. The customer fills in the vehicle problems in the input field.

15. The page ConfirmFormView.jsp sends a POST request to AppointmentServlet to request saving the appointment.
16. AppointmentServlet receives the request and calls the method doPost(request, response).
17. The class AppointmentDAO calls the method saveAppointment(Date date, Time time, String VehicleProblems).
18. The method saveAppointment() calls the class Appointment to encapsulate the appointment information.
19. The class Appointment encapsulates the entity information
20. Appointment returns the result to saveAppointment()
21. AppointmentDAO returns to AppointmentServlet.
22. AppointmentServlet returns a success response to ConfirmFormView.jsp
23. ConfirmFormView.jsp displays the message “Appointment saved successfully.”

Sequence diagram:



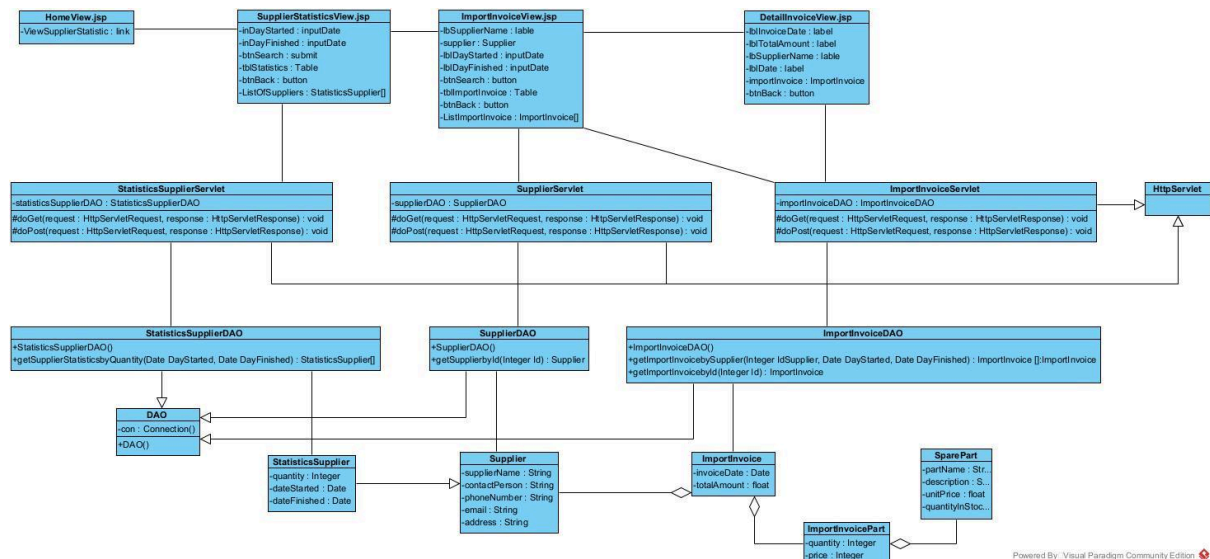
Module 2: Management staff views statistics of supplier

User interface is the jsp page:

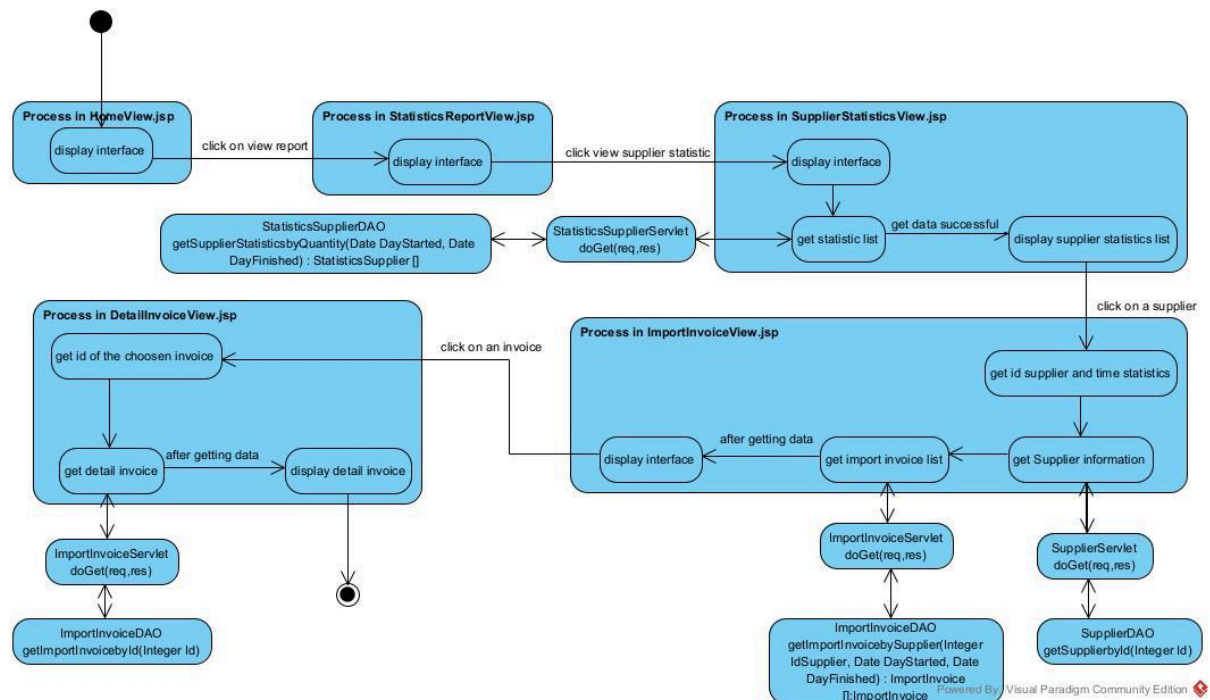
HomeView.jsp, StatisticsReportView.jsp, SupplierStatisticsView.jsp, ImportInvoiceView.jsp, DetailInvoiceView.jsp.

DAO for manipulate the model:

DAO, StatisticsSupplierDAO, SupplierDAO, ImportInvoiceDAO.



Activity Diagram:



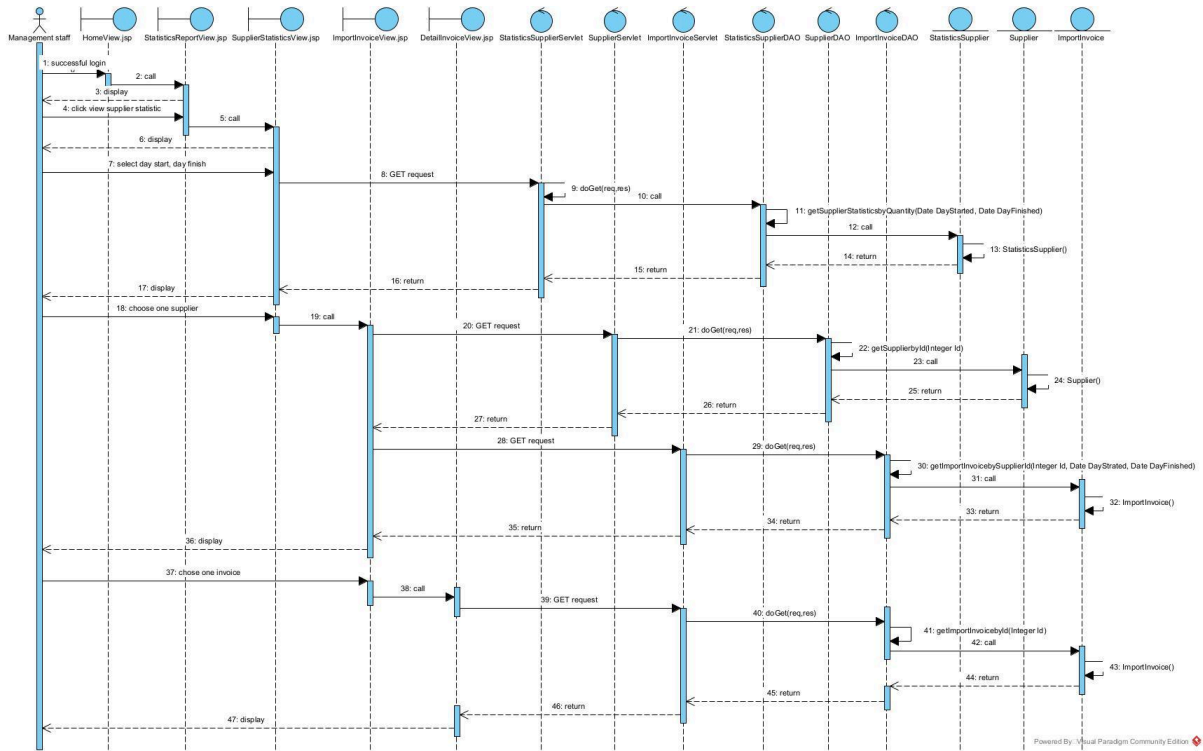
Scenario ver3:

1. On the main interface HomeView.jsp, after successful login, the management staff accesses the system.
2. The page HomeView.jsp calls the page StatisticsReportView.jsp.
3. The page StatisticsReportView.jsp displays the available report options.
4. The staff clicks View Supplier Statistics.
5. The page StatisticsReportView.jsp calls the page SupplierStatisticsView.jsp.
6. The page SupplierStatisticsView.jsp is displayed.

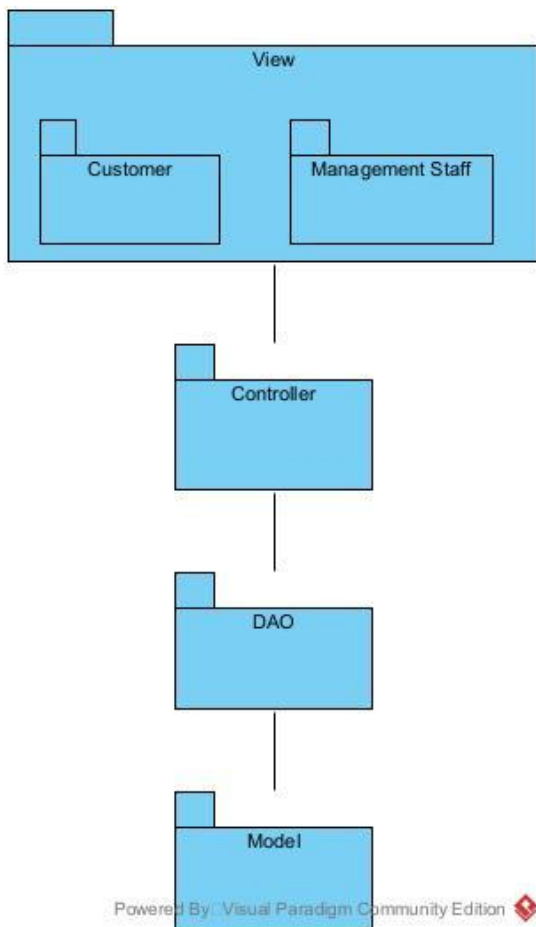
7. The management staff selects the start date and finish date for the report.
8. The page SupplierStatisticsView.jsp sends a GET request to StatisticsSupplierServlet to request supplier statistics within the selected period.
9. StatisticsSupplierServlet receives the request and executes doGet(request, response).
10. Inside doGet(), StatisticsSupplierServlet calls StatisticsSupplierDAO to request supplier statistics.
11. The class StatisticsSupplierDAO calls the method getSupplierStatisticsByQuantity(Date DayStarted, Date DayFinished).
12. The method getSupplierStatisticsByQuantity() calls the class StatisticsSupplier to encapsulate supplier statistics data.
13. The class StatisticsSupplier encapsulates the entity information.
14. The class StatisticsSupplier returns the result to getSupplierStatisticsByQuantity().
15. The method getSupplierStatisticsByQuantity() returns the result to StatisticsSupplierDAO, then to StatisticsSupplierServlet.
16. StatisticsSupplierServlet returns the response to SupplierStatisticsView.jsp.
17. The page SupplierStatisticsView.jsp displays the supplier statistics according to the selected period.
18. The management staff selects one supplier to view details.
19. The page SupplierStatisticsView.jsp calls the page ImportInvoiceView.jsp.
20. The page ImportInvoiceView.jsp sends a GET request to SupplierServlet to request supplier information.
21. SupplierServlet receives the request and executes doGet(request, response).
22. Inside doGet(), SupplierServlet calls SupplierDAO to request supplier information.
23. The class SupplierDAO calls the method getSupplierById(Integer Id).
24. The method getSupplierById() calls the class Supplier to encapsulate supplier data.
25. The class Supplier encapsulates the entity information.
26. The class Supplier returns the result to getSupplierById().
27. The method getSupplierById() returns the result to SupplierDAO, then to SupplierServlet.

28. SupplierServlet returns the response to ImportInvoiceView.jsp.
29. The page ImportInvoiceView.jsp sends a GET request to ImportInvoiceServlet to request import invoices of the selected supplier.
30. ImportInvoiceServlet receives the request and executes doGet(request, response).
31. Inside doGet(), ImportInvoiceServlet calls ImportInvoiceDAO to request supplier's import invoices.
32. The class ImportInvoiceDAO calls the method getImportInvoiceBySupplier(Integer IdSupplier, Date DayStarted, Date DayFinished).
33. The method getImportInvoiceBySupplier() calls the class ImportInvoice to encapsulate import invoice data.
34. The class ImportInvoice encapsulates the entity information.
35. The class ImportInvoice returns the result to getImportInvoiceBySupplier().
36. The method getImportInvoiceBySupplier() returns the result to ImportInvoiceDAO, then to ImportInvoiceServlet.
37. ImportInvoiceServlet returns the response to ImportInvoiceView.jsp.
38. The page ImportInvoiceView.jsp displays the list of import invoices.
39. The management staff selects one invoice to view details.
40. The page ImportInvoiceView.jsp calls the page DetailInvoiceView.jsp.
41. The page DetailInvoiceView.jsp sends a GET request to ImportInvoiceServlet to request details of the selected invoice.
42. ImportInvoiceServlet receives the request and executes doGet(request, response).
43. Inside doGet(), ImportInvoiceServlet calls ImportInvoiceDAO to get invoice details.
44. The class ImportInvoiceDAO calls the method getImportInvoiceById(Integer Id).
45. The method getImportInvoiceById() calls the class ImportInvoice to encapsulate invoice detail data.
46. The class ImportInvoice returns the result to getImportInvoiceById(), then to ImportInvoiceDAO, then to ImportInvoiceServlet.
47. ImportInvoiceServlet returns the response to DetailInvoiceView.jsp, which displays the detailed information of the selected invoice.

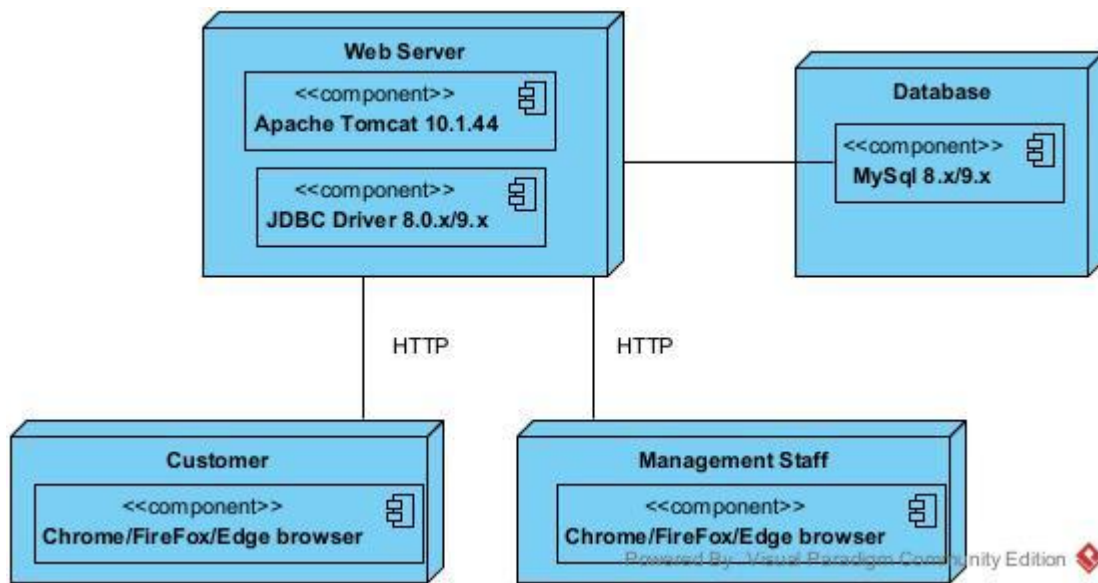
Sequence diagram:



4. Package diagram:

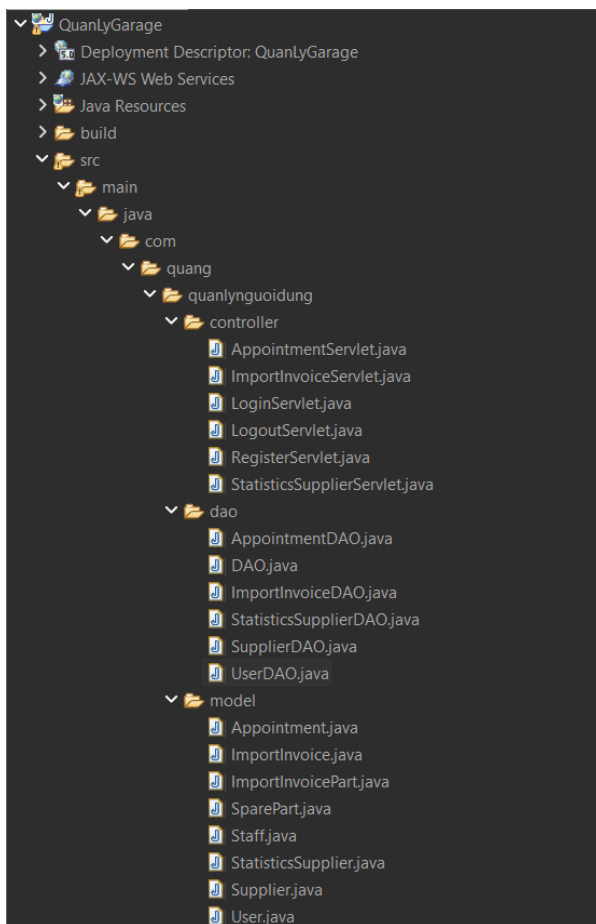


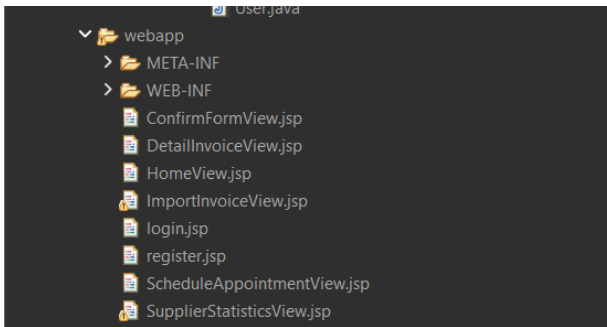
5. Deployment diagram:



D. Programming:

1. Project structure:





2. User interface:

Login:

A screenshot of a web application's login page. The page has a light gray background. In the center, there is a white card with a blue border and a shadow. The card is titled 'User Login' in bold black text. Below the title, there are two input fields: 'Username:' and 'Password:'. Each input field has a white background and a light gray border. Below the password field, there is a blue button with the text 'Login' in white. At the bottom of the card, there is a link that says 'Chưa có tài khoản? Đăng ký tại đây' in blue text.

Home:

A screenshot of a web application's home page. The page has a light gray background. In the center, there is a white card with a blue border and a shadow. The card is titled 'Garage Management System' in bold black text. Below the title, there is a welcome message 'Welcome, test01!' in blue text, followed by 'Your role is: user' in black text. Below this, there are three buttons: 'Schedule Repair Appointment' (blue), 'View Supplier Statistics' (blue), and 'Logout' (gray). The buttons are arranged vertically and have a slight shadow.

ScheduleAppointment:

Schedule Appointment

Select Date:

/ dd / yyyy

Select Time:

-- : -- --

Next

[Back to Home Page](#)

ConfirmForm:

Confirm Information

Customer: test03

Email: test03@gmail.com

Date of Birth: 2003-03-03

Phone Number: 0303003300

Appointment Date: 2025-11-18

Appointment Time: 13:20

Enter reason/vehicle issue:

bảo dưỡng

Confirm Booking

[Back](#)

SupplierStatistic:

Supplier Statistics

From Date:

10 / 01 / 2025

To Date:

11 / 30 / 2025

View Statistics (btnSearch)

Statistical Results

Supplier	Order Count	Total Amount	Details
AutoTech Supplies Co.	4	15010000 VNĐ	View Orders
SpeedParts Vietnam	3	13440000 VNĐ	View Orders
ProGarage Imports	3	10960000 VNĐ	View Orders
MegaAuto Components	2	8290000 VNĐ	View Orders
MotorCare Distribution	3	6000000 VNĐ	View Orders

Back to Home Page

ImportInvoice:

Invoices

Supplier:

AutoTech Supplies Co.

From Date:

2025-10-01

To Date:

2025-11-30

Invoice ID	Date Created	Total Amount	Details
HD-14	2025-11-08	6840000 VNĐ	View
HD-9	2025-10-25	2670000 VNĐ	View
HD-6	2025-10-18	2500000 VNĐ	View
HD-1	2025-10-01	3000000 VNĐ	View

Back (btnBack)

Link github:

https://github.com/qwerty12321434/PTTKHT_garage_management